

EXACT HISTORY DYNAMICS, DOMAIN-SAFE MEMORY, AND UNIVERSAL EXCESS PATH PRESSURE

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ABSTRACT. Conditional kernels are defined on a stable-leaf quotient history, never on a complete microscopic state whose future is deterministic. A renewal formula constructs the continuous-time Feller semigroup before Ray compactification. The history Doob eigenfunction is defined without a last-return ambiguity. Finite transmission zeros are extracted as actual Riesz residue modes of the compressed resolvent, giving an operator-compatible Schur dilation and a zero-free residual memory. Enhanced conditional homogenization retains the deterministic area anomaly.

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CONTENTS

1. Round-seven reconstruction: coarse histories, suspension kernels, and a Riesz–Schur memory dilation	1
1.1. Full and coarse histories	2
1.2. A continuous-time suspension semigroup	2
1.3. History eigenfunctions and the Doob transform	3
1.4. Riesz–Schur dilation of transmission zeros	3
1.5. Enhanced and conditional homogenization	5
Scope and verification status	5

1. ROUND-SEVEN RECONSTRUCTION: COARSE HISTORIES, SUSPENSION KERNELS, AND A RIESZ–SCHUR MEMORY DILATION

This section replaces the round-six controlling module. The full microscopic Sinai state is never used as the conditioning variable for a nondegenerate future law. Conditional kernels live on a stable-leaf quotient which forgets future-determining coordinates. Continuous-time kernels are constructed before Ray theory. Transmission zeros are treated by finite-rank Riesz residues of the compressed resolvent, not by a rational Smith–McMillan claim for a general meromorphic matrix.

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1.1. Full and coarse histories. Let E_{mic} be the graph-completed suspension state and let Φ_t^ε be its deterministic invertible flow. Let

$$\pi_s : E_{\text{mic}} \longrightarrow E_s$$

be the quotient by local stable leaves together with the present slow coordinate and the suspension clock. Conditional measures on stable fibers are the normalized Gibbs disintegrations λ_y^ε . The coarse history space is

$$H_s = D((-\infty, 0], E_s), \quad \mathcal{F}_t^s = \sigma\{\pi_s(\Phi_r^\varepsilon x) : r \leq t\}.$$

The full microscopic history still determines a Dirac future; no theorem below claims otherwise.

For a coarse history h ending at $y = h(0)$ define

$$K_t^\varepsilon(h, A) = \int_{\pi_s^{-1}(y)} 1_A(\pi_s(\Phi_{[0,t]}^\varepsilon x)) d\lambda_h^\varepsilon(x),$$

where λ_h^ε is the stable-fiber conditional law obtained by transporting the quotient g -kernel along the past. Exponential memory loss makes this version independent of remote representatives of h .

Lemma 1.1 (Nondegenerate coarse disintegration). *For every regular coarse history, λ_h^ε is a probability on a positive-dimensional stable fiber. The kernel K_t^ε is a Dirac mass only when the quotient fiber itself is a singleton. Given the complete microscopic history, the conditional kernel is explicitly the Dirac mass at the deterministic future, and that kernel is not used in the homogenization statement.*

Proof. The local product structure disintegrates the prepared Gibbs measure into unstable quotient and stable conditional measures with Holder Jacobians. The coarse past fixes the quotient itinerary but not the stable coordinate. The usual contraction of stable holonomies makes the conditional density a Cauchy limit as the remote past is extended. Conditioning on the full state instead fixes the stable coordinate, yielding the stated Dirac law. $\square$

1.2. A continuous-time suspension semigroup. Let P be the one-step quotient g -kernel, τ the roof, and $N_t(x)$ the number of returns completed by time t . For bounded uniformly continuous F on E_s , define for every real $t \geq 0$

$$(\mathcal{P}_t F)(y, u) = \sum_{n \geq 0} \int F(T^n x, u + t - S_n \tau(x)) 1_{\{S_n \tau(x) \leq u + t < S_{n+1} \tau(x)\}} dP_y^{(n)}(x).$$

This renewal formula defines irrational and rational times simultaneously.

Theorem 1.2 (Coarse-history Feller semigroup). *The operators $\mathcal{P}_t$ form a conservative strongly continuous Feller semigroup on the weighted uniformly continuous history state. Their kernels satisfy Chapman–Kolmogorov for all real $s, t \geq 0$. The resolvent*

$$R_\lambda F = \int_0^\infty e^{-\lambda t} \mathcal{P}_t F dt$$

is therefore a genuine sub-Markov resolvent satisfying the resolvent identity before any Ray compactification is invoked.

Proof. Partitioning according to N_t gives an absolutely convergent sum because the roof is bounded below and has an exponential tail. Splitting a path at time s and then at time $s + t$ proves Chapman–Kolmogorov term by term. Stable holonomy continuity and the roof tail give the Feller modulus; dominated convergence proves strong continuity at zero. Fubini and the semigroup identity yield the resolvent identity. $\square$

Choose a countable Ray cone generated by $R_\lambda F$ for a separating family of F and rational $\lambda > 0$.

Corollary 1.3 (Noncircular Ray realization). *The Ray compactification of the resolvent in Theorem 1.2 carries a right-continuous strong Markov process whose transition semigroup extends $\mathcal{P}_t$. The construction does not use the desired right continuity to define the resolvent.*

Proof. The already constructed resolvent is conservative, separates points on the Ray cone, and is normal by monotone convergence. The standard Ray theorem applies. $\square$

1.3. History eigenfunctions and the Doob transform. Let L_Ψ be the quotient transfer operator for a bounded source Ψ . On each positive recurrent component let h_Ψ and ℓ_Ψ be its strictly positive eigenfunction and eigenfunctional, normalized by $\ell_\Psi(h_\Psi) = 1$. Lift h_Ψ to the suspension by the renewal formula rather than by a possibly nonexistent last return:

$$H_\Psi(y, u) = e^{-\lambda_\Psi u} \mathbb{E}_y \left[e^{\int_0^{T_1 - u} \Psi(X_s) ds} h_\Psi(Y_1) \right].$$

This formula is defined at every coding representative and has matching oriented traces.

Theorem 1.4 (Bounded history Doob transform). *On compact source charts, $0 < c \leq H_\Psi \leq C < \infty$. The operators*

$$\mathcal{P}_t^\Psi F = e^{-t\lambda_\Psi} H_\Psi^{-1} \mathcal{T}_t^\Psi (H_\Psi F)$$

form a conservative Feller semigroup on the coarse history state and preserve the Doob invariant probability.

Proof. The quotient cone theorem gives uniform upper and lower bounds for h_Ψ on the normalized inducing base. The roof exponential moment and bounded source transfer these bounds through the renewal formula. The eigenfunction identity proves conservation and the semigroup law. $\square$

1.4. Riesz–Schur dilation of transmission zeros. Let L_D be the generator of one fixed Doob semigroup on $\mathcal{H} = L^2(\nu_D)$, let P be the orthogonal projection onto a finite resolved space $V \subset D(L_D^r)$, and define

$$C(z) = P(z - L_D)^{-1} P|_V.$$

Assume the A2 resolvent estimate in a strip $\operatorname{Re} z > -\gamma$ and let $\mathcal{Z}_0$ be the finite multiset of zeros of $\det C(z)$ in the smaller strip $\operatorname{Re} z \geq -\gamma_0$. Around each zero take a contour Γ_j avoiding poles and define the finite-rank residue space generated by the Laurent coefficients of $C(z)^{-1}$.

Theorem 1.5 (Operator-compatible Riesz–Schur memory dilation). *There exist a finite-dimensional space Z , matrices A_Z, B_Z, C_Z , and an orthogonal projection $\tilde{P}$ on $V \oplus Z$ such that the Schur complement of*

$$\tilde{C}(z) = \tilde{P}(z - \tilde{L})^{-1}\tilde{P}$$

with respect to Z equals $C(z)$ on $\operatorname{Re} z > 0$. The spectrum of A_Z consists exactly of the transmission zeros in $\mathcal{Z}_0$, with their algebraic multiplicities. The residual compressed inverse is analytic and zero-free in $\operatorname{Re} z \geq -\gamma_0$.

Proof. The analytic Fredholm theorem makes $C(z)^{-1}$ meromorphic in the strip. Its principal parts at the finitely many zeros are finite-rank matrix polynomials in $(z - z_j)^{-1}$. Realize each principal part by its companion Jordan block on a finite residue space and take their direct sum. Subtracting these principal parts leaves an analytic inverse. The block operator

$$\tilde{L} = \begin{pmatrix} PL_D P & C_Z \\ B_Z & A_Z \end{pmatrix}$$

with the unresolved Q -resolvent retained in the V block has, by the block resolvent identity, Schur complement equal to the original compressed resolvent. The coefficients were extracted from the actual Laurent residues, so this is an operator-compatible dilation rather than an arbitrary rational factorization. The remaining inverse is zero-free by construction. $\square$

Let

$$\widehat{K}_{\text{res}}(z) = C_{\text{res}}(z)^{-1} - zI - A_{\text{res}}.$$

Theorem 1.6 (Residual memory decay). *If $V \subset D(L_D^r)$ with r larger than the polynomial loss in the A2 high-frequency resolvent by two, then*

$$\|K_{\text{res}}(t)\| \leq Ce^{-\gamma_0 t}.$$

The exact resolved equation is the coupled Volterra system on $V \oplus Z$; all slow auxiliary modes and the residual memory are obtained from the same microscopic time scaling.

Proof. Theorem 1.5 removes every zero and pole in the contour-shift strip. The $D(L_D^r)$ assumption gives a vertical-line bound integrable after r integrations by parts. Shift the Bromwich contour to $\operatorname{Re} z = -\gamma_0$ and estimate the horizontal sides using the A2 Dolgopyat loss. The companion modes are part of the same accelerated resolvent, so their scaling follows from the pole locations rather than being inserted as an unscaled ODE. $\square$

1.5. Enhanced and conditional homogenization. Assume the induced observable has exponential moments. Then martingale–coboundary decomposition supplies moments of every finite order for first and second levels. Let X^ε denote the slow observable and $\mathbb{X}^\varepsilon$ its canonical iterated integral.

Theorem 1.7 (Enhanced coarse-history invariance principle). *For every $p > 2$, $(X^\varepsilon, \mathbb{X}^\varepsilon)$ converges in p -variation rough-path topology to the diffusion rough path with Green–Kubo covariance and the deterministic area anomaly. Moreover, uniformly on compact subsets of $\mathbf{H}_s$,*

$$K_T^\varepsilon(h, \cdot) \implies K_T^{\text{diff}}(X_0(h), \cdot).$$

This statement is conditional on the coarse quotient history. Conditional on the full microscopic history the kernel remains Dirac.

Proof. The all-order moment bounds give Kolmogorov estimates for increments and the second level, hence tightness for every $p > 2$. Finite-dimensional convergence follows from the martingale approximation; antisymmetric coboundary terms give the area anomaly. For conditional convergence, truncate the coarse past at a Young return. Stable-fiber coupling makes the truncation error exponentially small uniformly on compact history sets. The finite-past kernel is an integral against λ_h^ε , so the same martingale estimates hold uniformly. Let the truncation recede after taking the homogenization limit. No deterministic point kernel is approximated by a Brownian law. $\square$

Theorem 1.8 (Round-seven A4 closure). *The conditional theory uses a genuine coarse history with stable-fiber randomness, possesses a continuous-time Feller semigroup and noncircular Ray resolvent, a globally defined history Doob eigenfunction, an operator-compatible Riesz–Schur dilation of transmission zeros, exponential residual memory under quantified generator regularity, and an enhanced conditional invariance principle consistent with determinism.*

Proof. Combine Lemma 1.1, Theorems 1.2, 1.4, 1.5, 1.6, and 1.7, and Corollary 1.3. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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